2 **(a)** Explain what is meant by *escape speed*.

.....

..... [1]

(b) A planet has radius R and the acceleration of free fall at its surface is g. The planet may be considered to be a sphere with its mass concentrated at its centre.

Deduce that the escape speed v_{es} is given by the expression

$$v_{es} = \sqrt{2gR}$$

Explain your working and state **one** assumption that is made in the derivation.

Assumption: .....

..... [3]

(c) Calculate the escape speed for a spherical planet of radius 1.7×10^3 km having an acceleration of free fall of $1.6\ \text{m s}^{-2}$ at its surface.

speed = ... m s^{-1} [2]

(d) The mean translational kinetic energy E_K of a helium-4 atom at thermodynamic temperature T is given by the expression

$$E_K = \frac{3}{2}kT$$

where k is Boltzmann's constant.

Determine the surface temperature of the planet such that helium-4 atoms on the surface of the planet are able to reach the escape speed calculated in **(c)**.

temperature = ... K [2]

(e) Suggest **one** reason why, at temperatures below that calculated in **(d)**, helium atoms can still escape from the planet.

.....

.....

..... [1]